CLAIMS

What is claimed is:

1. A satellite communication system, comprising:

    - a ground station, comprising:
        - a high-radix RF modulator configured to receive a sequence of abstract symbols and generate a corresponding sequence of N discrete, distinguishable radio frequency (RF) states, where N is an integer greater than two;
        - an antenna for transmitting said sequence of RF states towards a satellite; and
    - a satellite transponder, comprising:
        - an antenna for receiving said sequence of RF states;
        - a high-radix RF transceiver configured to process said sequence of RF states in its symbolic form; and
        - an antenna for retransmitting said sequence of RF states towards a receiving ground station;
    - a receiving ground station, comprising:
        - an antenna for receiving said retransmitted sequence of RF states; and
        - a high-radix RF demodulator configured to detect and distinguish the N discrete RF states and generate a corresponding sequence of abstract symbols.

2. The system of claim 1, further comprising an adaptive link manager configured to compensate for satellite channel impairments, including Doppler shift, by dynamically adjusting the definition or mapping of the N RF states.

3. The system of claim 1, wherein the abstract symbols are components of a parametric data packet, the payload of which is an unevaluated mathematical expression.

4. The system of claim 3, wherein the high-radix RF modulator is configured to directly modulate the parametric packet's payload without intermediate binary conversion.

5. The system of claim 1, wherein the high-radix RF transceiver on the satellite transponder processes the data in its symbolic form without converting it to a binary representation.

6. A method for satellite communication, comprising:

   - at a transmitting ground station, receiving a data structure representing an abstract symbol;
   - generating, based on said abstract symbol, one of N discrete, distinguishable radio frequency (RF) states, where N is an integer greater than two;
   - transmitting said RF state towards a satellite;
   - at the satellite, receiving said RF state, processing it symbolically, and retransmitting it towards a receiving ground station;
   - at the receiving ground station, receiving said RF state; and
   - distinguishing said RF state from the other N-1 possible RF states and generating the original abstract symbol.

7. The method of claim 6, further comprising dynamically compensating for Doppler shift by adjusting the interpretation of the N RF states.

8. The method of claim 6, wherein the abstract symbols represent an unevaluated mathematical expression that is transported across the satellite link without loss of precision.